Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

Prepared by: Samia Nawsheen
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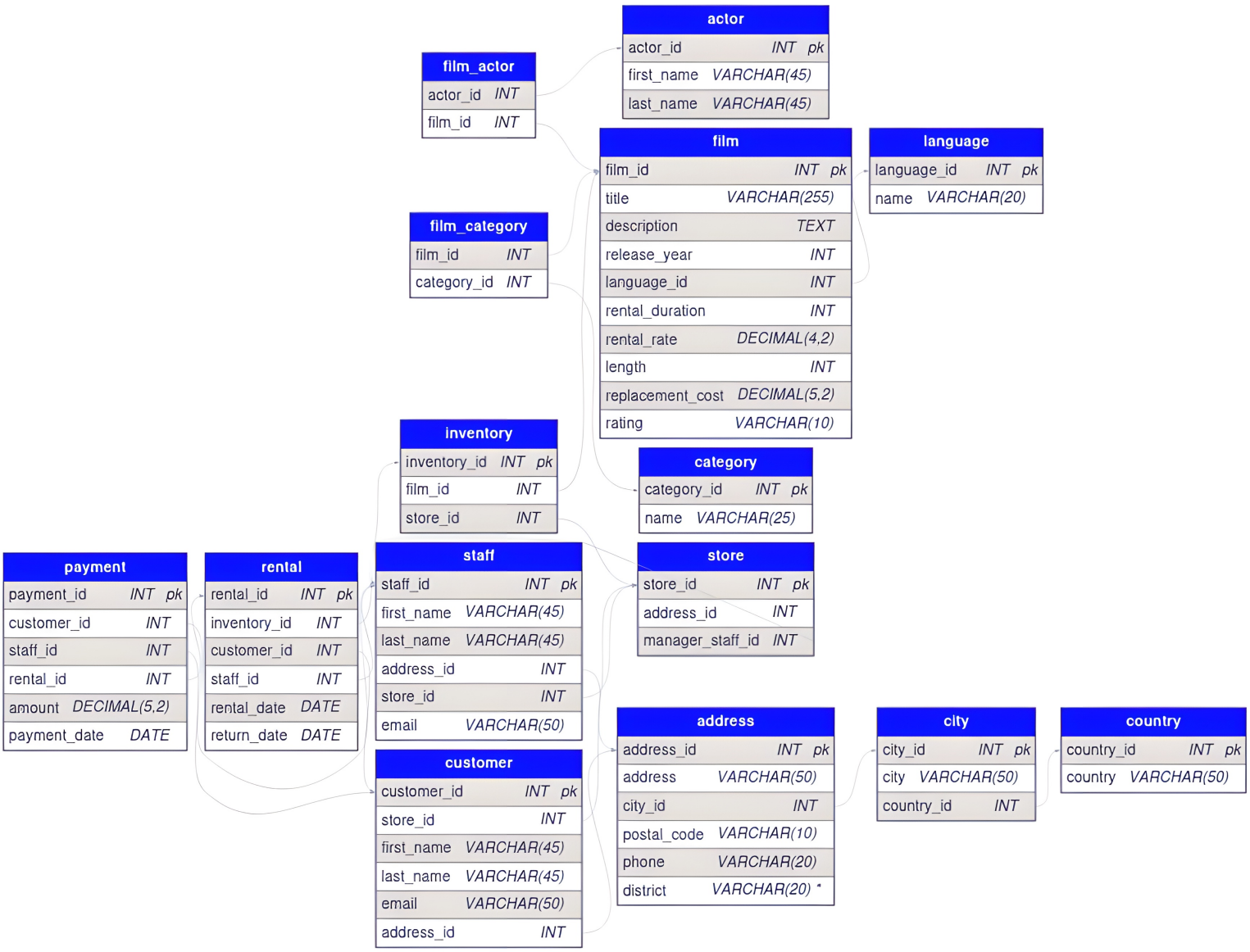
DBMS Lab Mid



1. Find the rentals that did not cross a calendar boundary between rent and return.

```
SELECT rental_id, rental_date, return_date
FROM rental
WHERE EXTRACT(MONTH FROM rental_date) =
      EXTRACT(MONTH FROM return_date);
```

2. List all customers whose email contains their first name.

   Solution:

```
SELECT first_name, last_name, email
FROM customer
WHERE POSITION(LOWER(first_name) IN LOWER(email)) > 0;
```

3. Show the total number of rentals that have been returned and the total number that have not been returned yet, as separate rows.

```
SELECT 'Returned' AS status, COUNT(*) AS total
FROM rental
WHERE return_date IS NOT NULL
```

```sql
UNION ALL

SELECT 'Not Returned' AS status, COUNT(*) AS total
FROM rental
WHERE return_date IS NULL;
```

4. Display the title of films that end with a consonant.

```sql
SELECT title
FROM film
WHERE LOWER(title) NOT LIKE '%a'
  AND LOWER(title) NOT LIKE '%e'
  AND LOWER(title) NOT LIKE '%i'
  AND LOWER(title) NOT LIKE '%o'
  AND LOWER(title) NOT LIKE '%u';
```

5. Find all customers whose first name starts with the letter 'A'.

```sql
SELECT first_name, last_name
FROM customer
WHERE first_name LIKE 'A%';
```

6. Identify the film categories that contain more than 50 films.

```sql
SELECT c.name, COUNT(fc.film_id) AS film_count
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
GROUP BY c.name
HAVING COUNT(fc.film_id) > 50;
```

7. Find out which customers have rented films from more than one store.

```sql
SELECT c.customer_id, c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN inventory i ON r.inventory_id = i.inventory_id
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING COUNT(DISTINCT i.store_id) > 1;
```

8. For each year, find the total number of rentals made in that year.

```sql
SELECT EXTRACT(YEAR FROM rental_date) AS year,
       COUNT(*) AS total_rentals
FROM rental
GROUP BY year
ORDER BY year;
```

9. Find customers who have rented films from ten different categories.

```sql
SELECT customer_id, first_name, last_name
FROM customer
WHERE customer_id IN (
    SELECT r.customer_id
    FROM rental r
    JOIN inventory i ON r.inventory_id = i.inventory_id
    JOIN film_category fc ON i.film_id = fc.film_id
    GROUP BY r.customer_id
    HAVING COUNT(DISTINCT fc.category_id) = 10
);
```

10. List the customers who have spent more money in total than the average total amount spent by a customer.

Solution:

```sql
SELECT
    c.first_name,
    c.last_name,
    SUM(p.amount) AS total_paid
FROM customer c
JOIN payment p ON c.customer_id = p.customer_id
GROUP BY c.customer_id
HAVING SUM(p.amount) > (
    -- Subquery: Calculate the Average of the Totals
    SELECT AVG(total_per_customer)
    FROM (
        -- Inner Subquery: Calculate Total per Customer first
        SELECT SUM(amount) AS total_per_customer
        FROM payment
        GROUP BY customer_id
    ) AS derived_table
);
```